Student Number

PEM

2025 **TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black pen.
- Draw diagrams using pencil.
- For questions in Section II, show all relevant working in questions involving calculations.
- NESA-approved calculators may be used.

Write your Student Number where required.

Total Marks – 100

Section I - 20 marks (pages 2 - 8)
- Attempt Questions 1 – 20.
- Allow about 35 minutes for this section.

Section II – 80 marks (pages 9 - 31)
- Attempt Questions 21 – 35.
- Allow about 2 hours and 25 minutes for this section.

Section I

20 marks

Attempt Questions 1 – 20.

Allow about 35 minutes for this section.

Use the multiple-choice answer sheet provided for Questions 1 – 20.

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1 A chemist mixed 2.0 mol of CO and 2.0 mol of Cl_2 in a 1 Litre vessel. The mixture was left to reach equilibrium at constant temperature. The equation for the reaction is:

$$CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)}$$

Once equilibrium was reached, the mixture contained 0.115 mol of Cl_2. Determine the value of K_{eq} for this equilibrium.

A. 143
B. 31
C. 0.025
D. 0.007

2 Consider a mixture that has reached equilibrium according to the equation:

$$2CO_{(g)} + 2NO_{(g)} \rightleftharpoons 2CO_{2(g)} + N_{2(g)} \qquad \Delta H = -788\ kJ.mol^{-1}$$

Which row of the following table correctly predicts the outcomes of the conditions imposed on the equilibrium mixture?

	Conditions imposed on equilibrium		**Outcomes**	
	Temperature	**Pressure**	**Reaction favoured**	**K_{eq} value**
A.	constant	increased	reverse	remains constant
B.	increased	constant	forward	increases
C.	constant	decreased	forward	decreases
D.	increased	decreased	reverse	decreases

3 The value of Q for a reversible reaction is determined by substituting the initial concentrations of reactants and products into the K_{eq} expression. If a gaseous mixture is prepared, the reaction that occurs will be:

A. the forward reaction if $Q < K_{eq}$.
B. the reverse reaction if $Q = K_{eq}$.
C. always in the direction that releases the most energy.
D. always in the direction that produces the fewest moles of gas.

4 The K_{sp} of lead (II) chloride at 25°C is 1.70×10^{-5}. What mass of lead chloride could you dissolve completely in 100 mL of water at 25°C?

A. 1.62×10^{-2} g
B. 4.50×10^{-1} g
C. 3.93×10^{-2} g
D. 1.15×10^{-1} g

5 Which of the following is an example of a non-reversible reaction?

A. Anhydrous cobalt (II) chloride turns pink when wet.
B. Copper sulphate pentahydrate turns white when heated.
C. Magnesium produces white light when heated.
D. Phenolphthalein indicator turns pink in basic solution.

6 In a 0.005 M HCl solution, the dissociation of water produces _____________M H_3O^+ and ____________M OH^-.

A. 5.00×10^{-3} M H_3O^+ and 5.00×10^{-3} M OH^-
B. 5.00×10^{-3} M H_3O^+ and 5.00×10^{-11} M OH^-
C. 1.00×10^{-7} M H_3O^+ and 1.00×10^{-7} M OH^-
D. 5.00×10^{-3} M H_3O^+ and 2.00×10^{-12} M OH^-

7 Which of the following is the conjugate base of a strong acid?

A. ClO^-
B. $H_2PO_4^-$
C. Cl^-
D. OH^-

The following two questions refer to the table below, which lists the pK_a of several monoprotic acids.

Acid	**pK_a**
benzoic	4.2
chlorous	1.9
hydrofluoric	3.2
nitrous	3.3
periodic	1.6

8 Which of the following statements is **CORRECT**?

A. Hydrofluoric acid is twice as strong as periodic acid.
B. Periodic acid is the weakest acid of those listed.
C. The dissociation of benzoic acid occurs more rapidly than the dissociation of chlorous acid.
D. The pK_a value is inversely related to the strength of the acid.

9 The pH of a 0.00023 M solution of benzoic acid is:

A. 3.9
B. 2.1
C. 2.3
D. 4.2

10 Which of the following compounds would react completely with excess $NaHCO_3$?

A. ethyl alcohol
B. ethanoic acid
C. ethanamine
D. none of the above

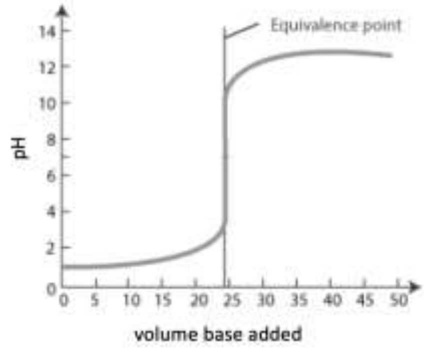
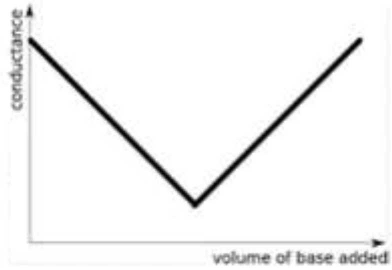
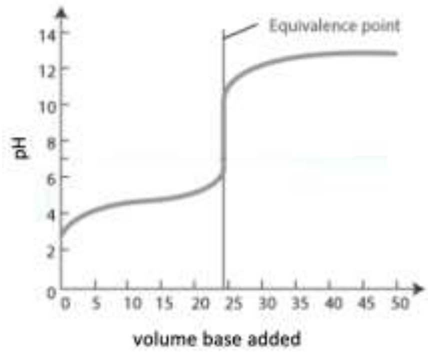
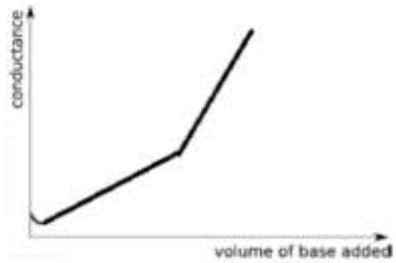
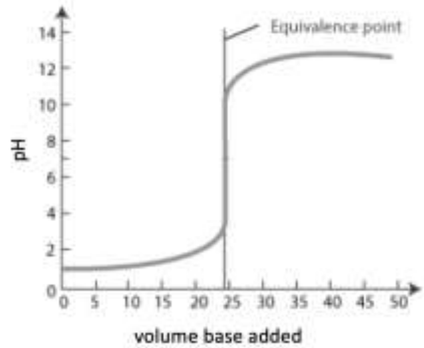
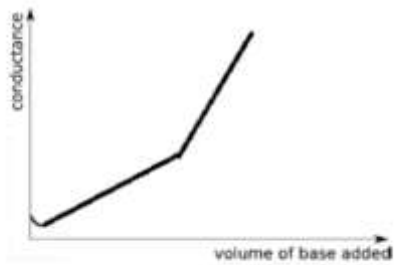
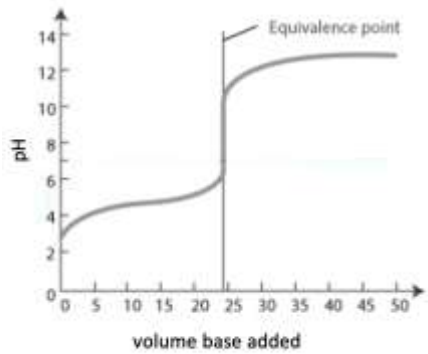
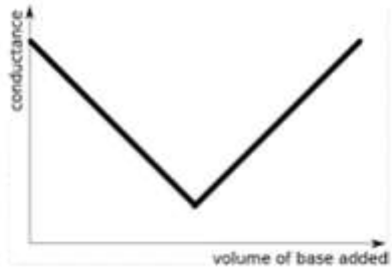
11 A student was attempting to identify the cation present in a clear solution. They tested separate samples of the solution with two different reagents and obtained the following results.

Added HCl	White precipitate
Added NH_3	White precipitate

Of the following four metals, the cation is most likely:

A. copper
B. calcium
C. magnesium
D. lead

12 Which of the following pairs of graphs represent the titration curve and conductivity curve for the titration of a weak acid with a strong base?

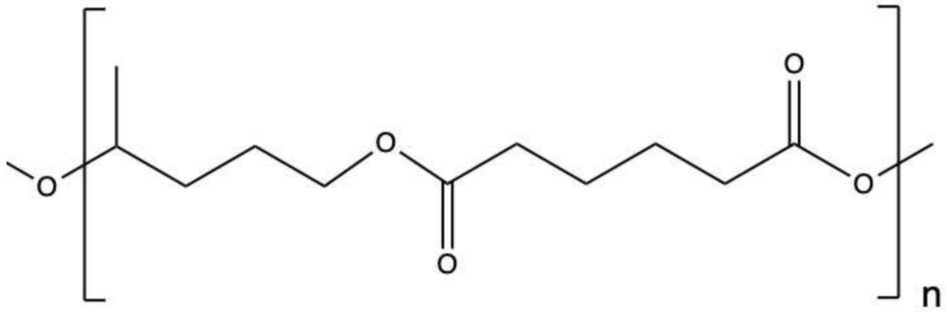
	Titration curve	**Conductivity curve**
A.		
B.		
C.		
D.		

13 Which of the following compounds would react with acidified potassium permanganate?

A. CH_3CH_2COOH
B. CH_3COCH_3
C. $CH_3CH_2CH_2O$
D. Both B and C

The following two questions refer to the information below.

In 2020, researchers in Germany published a paper outlining their development of a novel polyester synthesis process that used a renewable diol produced from plant material.[1] The process has the potential to replace some fossil-based plastics currently used in the production of medical devices, automotive coatings and adhesives. One such polyester is illustrated below.



14 Which of the following is the name of the diol used in the synthesis of this polymer?

A. 1,4-pentanediol
B. 2,5-hexanediol
C. 1-methylbutanediol
D. 1,6-hexanediol

15 The molecular mass of the polyester is 5800 $g.mol^{-1}$. How many diol monomers were incorporated into this polymer molecule?

A. 25
B. 27
C. 29
D. 31

[1] Stadler, BM *et al.* (2020) Properties of novel polyesters made from renewable… (*name of diol*). *ChemSusChem 13*: 556-563

16 In a practical examination, a student was asked to devise a technique to convert dichloroethane to ethan-1,2-diol in a single reaction. Which of the following reagents should the student use?

A. NaOH

B. Hydrogen with a palladium catalyst

C. CH_3COOH

D. H_2O/H^+

17 After carrying out the reaction to convert dichloroethane to ethan-1,2-diol, the student decided to test for the presence of chloride ions in the final solution. If the dichloroethane had been successfully converted to ethan-1,2-diol, there should be free chloride ions present in the final mixture. Which of the following reagents should the student use to test for the presence of free chloride ions?

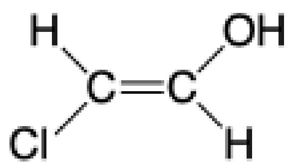
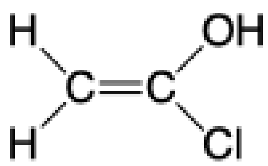
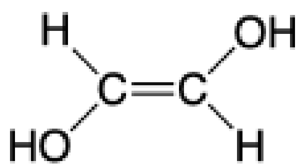
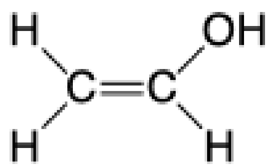
A. HCl

B. KNO_3

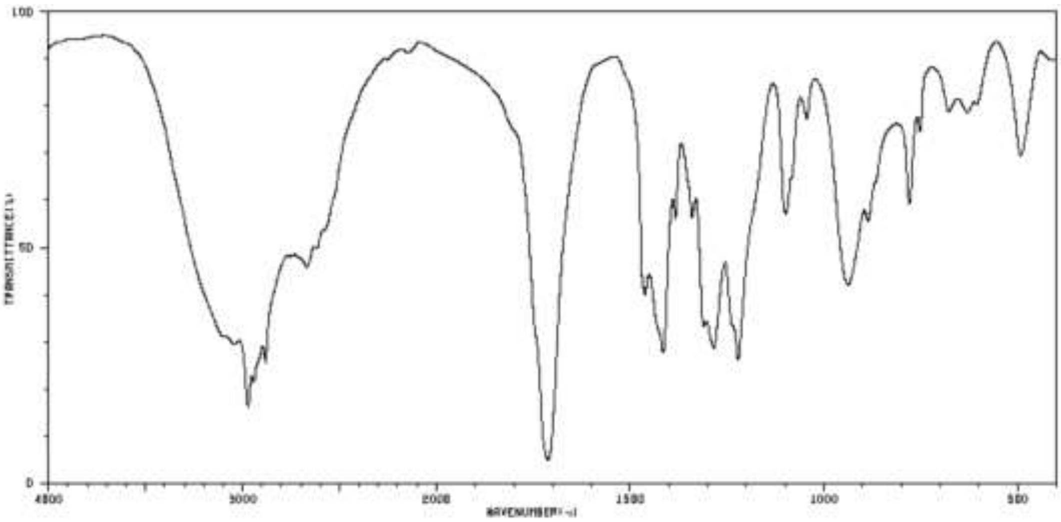
C. NaOH

D. $AgNO_3$

18 Which of the following compounds would produce the same number of signals on their ^{1}H-NMR and ^{13}C-NMR spectra?

A.		B.	
C.		D.	

19 Which of the compounds below is most likely to have produced the following IR spectrum?



A. butanone

B. butanol

C. butanoic acid

D. butane

20 Phosphorite is a mineral containing phosphate, hydroxide and calcium ions. A chemist wanted to determine the calcium content of an impure sample of phosphorite. A 4.64 g sample was dissolved then the calcium was precipitated out as insoluble calcium oxalate hydrate ($CaC_2O_4.H_2O$). The precipitate was filtered and heated at a temperature that was high enough to dry it while retaining the water of hydration in the crystals. The mass of the precipitate obtained was 2.86 g.

The mass percent of calcium in the phosphorite was:

A. 5.32%

B. 13.3%

C. 16.9%

D. 19.3%

2025 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Student Number

Section II Answer Booklet

80 marks

Attempt Questions 21 – 35.

Allow about 2 hours and 25 minutes for this section.

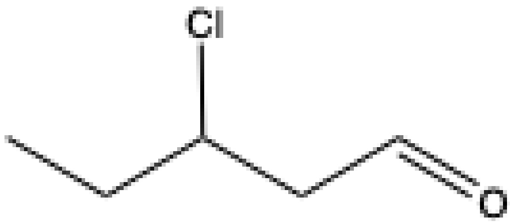
Instructions

- Write your Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over.

Question 21. (2 marks)

Draw and name one chain isomer and one functional group isomer of the following compound.



(2 marks)

Chain isomer	Functional group isomer
Name:	Name:...

Question 22. (3 marks)

A laboratory technician was asked to make up a buffer for a biochemistry experiment. They had available three different sodium salts: sodium phosphate, sodium hydrogen phosphate and sodium dihydrogen phosphate.

Choose two of the compounds and explain how, when mixed in equal proportions, they behave as a buffer. Use an appropriate chemical equation in your answer. *(3 marks)*

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Question 23. (4 marks)

Water is an amphoteric compound. Using appropriate definitions and equations, explain how water can act as both a Brønsted-Lowry acid and a Brønsted-Lowry base. *(4 marks)*

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Question 24. (4 marks)

A popular cleaning product sold online is made of water and potassium hydroxide. The concentration of potassium hydroxide is advertised as less than 0.5% (= 0.5 g/100 mL).

(a) Calculate the pH of this product, assuming the concentration of KOH is equal to 0.5%.

(1 mark)

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(b) A student decided to determine the actual concentration of KOH in the cleaning product by carrying out a titration. They used 50 mL aliquots of the cleaning product in a conical flask and found that an average of 16.2 mL of 0.250 M HCl was required to neutralise the KOH present.

Calculate the concentration of KOH in the product in mol.L^{-1} and % (g/100 mL).

(3 marks)

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Question 25. (9 marks)

A laboratory technician followed the procedure below to make up a 0.2% solution of methyl red indicator.

"Weigh out 1.00 g of methyl red powder (molar mass 269.3 $g.mol^{-1}$) on an analytical balance. Place the powder into a 500 mL volumetric flask and add 100 mL of 95% ethanol. Mix until dissolved. Add sufficient ethanol to make up to 500 mL."

(a) Calculate the molarity of the 0.2% methyl red solution. *(1 mark)*

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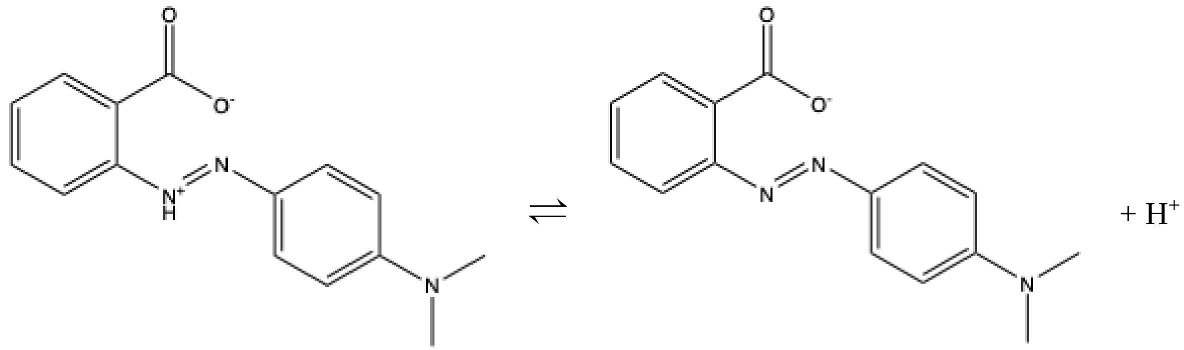
(b) The non-ionised form of methyl red is red and the anion is yellow. The structures of the non-ionised form and the anion are given below.



acidic (red) basic (yellow)

The equilibrium between the two forms can be summarised as:

$$MRH \rightleftharpoons MR^- + H^+$$

The pK_a of methyl red is 4.95. Determine the pH of a 0.2% solution in which the methyl red is:

(i) 50% ionised. *(1 mark)*

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Question continues over page.

(ii) 90% ionised. *(2 marks)*

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(c) A student wished to carry out a titration between NaOH and acetic acid solution. Would methyl red be a suitable indicator to use in this titration? Explain your answer. *(2 marks)*

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(d) The Beer-Lambert Law indicates the relationship between the absorbance of light and the concentration of a solution. The equation is:

$$A = \varepsilon \ell c$$

where A = absorbance

ε = the molar absorption coefficient (molar absorptivity) of the compound

ℓ = the path length of the sample

c = the concentration (molarity) of the compound in solution.

A solution of methyl red at pH 1 has an absorbance of 0.315 at 515 nm in a 2.00 mm cell. The molar absorptivity of methyl red at 515 nm is 2.30×10^4 $L.mol^{-1}.cm^{-1}$.

Calculate the molarity of methyl red in this solution. *(1 mark)*

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(e) The following graphs indicate the absorbance of a standard solution of methyl red at various pH.

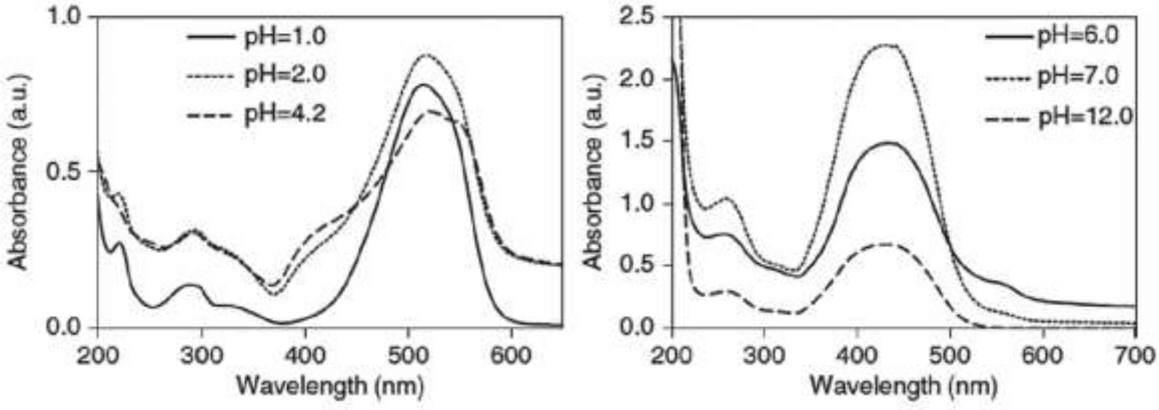

Figure 1. UV-visible spectra of methyl red at various pH.[2]

What wavelength of light would be best to use for carrying out a colourimetric analysis of methyl red in basic solution? Explain your answer. *(2 marks)*

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2 Thomas, O. and Brogat, M (2022) Organic Constituents. (In) *UV-Visible Spectrophotometry of Waters and Soils (Third Edition)* pp 95-160 (Elsevier: Netherlands)

Question 26. (6 marks)

An experiment was carried out to determine the molar enthalpy of neutralisation for the reaction between 1 M $NaHCO_3$ and 1 M HCl.

75 mL of 1 M HCl was added to 75 mL of 1 M $NaHCO_3$ in a calorimeter and the gas evolved was collected. The temperature of the mixture changed from 22°C to 29° C during the reaction.

(a) Write a balanced equation for the reaction. *(1 mark)*

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(b) Calculate the molar enthalpy of neutralisation for the reaction.
(Assume the density and heat capacity of the mixture is the same as that of water and the heat gained by the calorimeter is negligible.) *(3 marks)*

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(c) Calculate the volume of the gas evolved (at 25°C) during the reaction. *(2 marks)*

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Question 27. (4 marks)

Solid NH_4HS is placed in an evacuated container at 25ºC and the following equilibrium is established. The total gas pressure in the chamber before the NH_4HS is added is zero.

$$NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)} \qquad \Delta H = +93 \text{ kJ.mol}^{-1}$$

At equilibrium, some solid NH_4HS remains in the container.

(a) Predict and explain the effect on the equilibrium partial pressure of NH_3 gas when additional solid $NH_4HS_{(s)}$ is introduced into the container. *(2 marks)*

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(b) Predict and explain the effect on the amount of solid NH_4HS present when the temperature is increased. *(2 marks)*

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Question 28. (9 marks)

Australia produces approximately 4.5 million tonnes of sugar annually, mostly from sugarcane harvested in Queensland. A by-product of the refining of sugar is bagasse, which is the fibrous component of the sugarcane. This material is usually burned to produce energy to power the sugar refinery. However, bagasse has more recently been used in processes that aim to replace fossil-based aviation fuels and the plastics used to make disposable biodegradable plates and cutlery.

In 2021, a pilot plant to create aviation fuel from bagasse was set up in a joint venture between Mercurius Australia and the Queensland University of Technology. The patented design for their biorefining process is summarised below.[3]

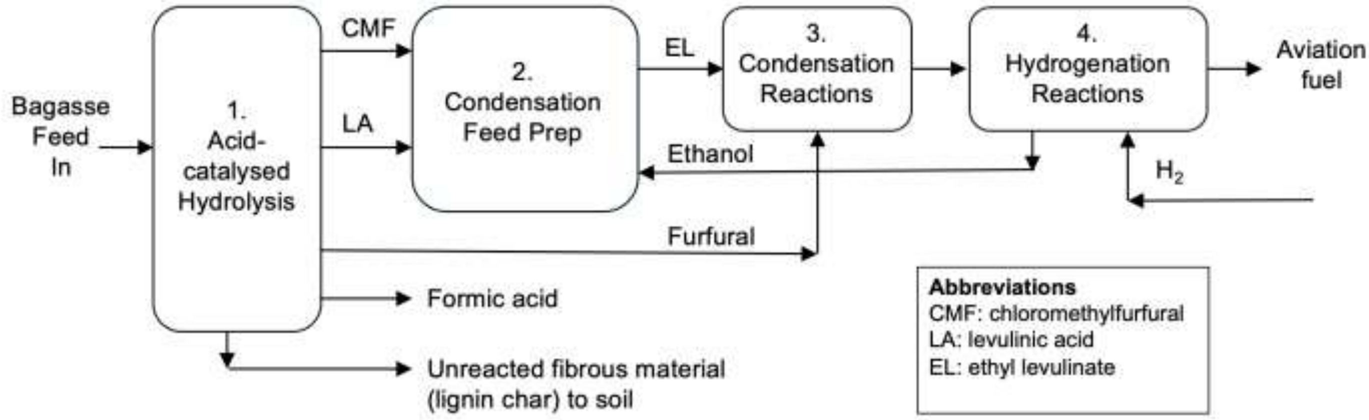

Figure 2. REACH (Renewable Acid-hydrolysis Condensation Hydrotreating) process for converting bagasse into aviation fuel.

(a) What is the role of acid-catalysed hydrolysis in the REACH process? *(1 mark)*

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Question continues over page.

3 Seck, K.A. (2019) Biorefinery for conversion of carbohydrates and lignocellulosics via primary hydrolysate CMF to liquid fuel. US Patent Office No.:10,174,262 B2

(b) Suggest two factors that you would need to consider when deciding on the location of a plant for producing aviation fuel from bagasse. *(2 marks)*

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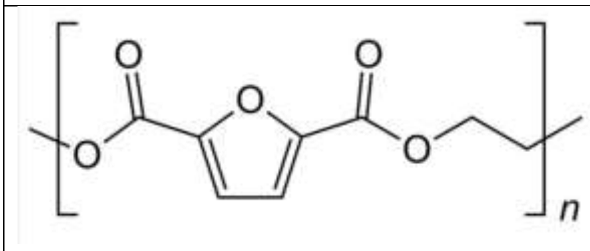
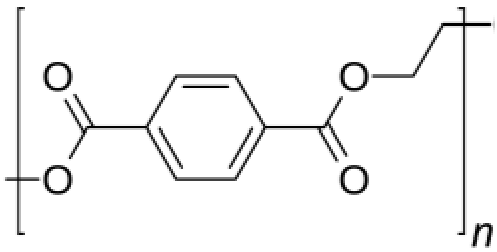
(c) Intermediate compounds from the REACH process have been used to produce a plastic called polyethylene furanoate (PEF). PEF has many advantages over the widely-used polyethylene terephthalate (PET), which is produced primarily from petroleum: it is a superior barrier against gases, is mechanically stronger and has better thermal stability.

The structures of the two polymers are illustrated below.

PEF	PET
	

Use your knowledge of polymer structures and properties to suggest why PEF produces a stronger and more thermally-stable plastic than PET. *(2 marks)*

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Question continues over page.

(d) Discuss the economic and environmental advantages of using bagasse in the REACH process to make aviation fuel and plastics. *(4 marks)*

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Question 29. (3 marks)

A student is presented with a clear solution, containing the sodium salt of an anion. The anion could be sulphate, acetate or iodide.

Outline a sequence of precipitation tests that could be carried out in a school laboratory to determine the identity of the anion. Include your expected observations and write balanced net ionic equations for all positive tests described in your answer. *(3 marks)*

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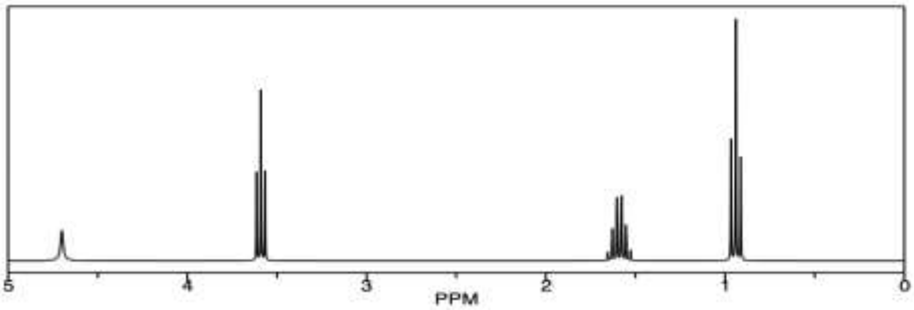
Question 30. (9 marks)

A compound (A) with empirical formula CH_2 was reacted with acidified water and produced a mixture of two products (B and C). The two products were separated to produce pure samples of each. Product B was treated with an oxidising agent and the resultant compound was purified. This newly-synthesised compound (D) was then reacted under suitable conditions with the second product of the initial reaction (product C) to produce compound E.

The ^{1}H-NMR and ^{13}C-NMR spectra of compounds B, C, D and E, and the mass spectrum of the original compound A, are given below.

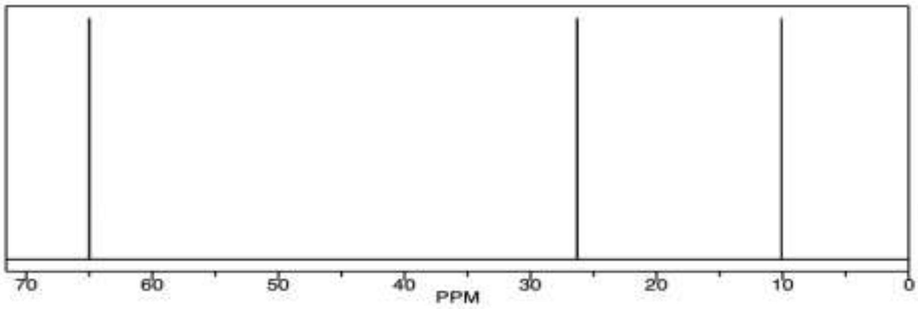
Product B

^{1}H-NMR



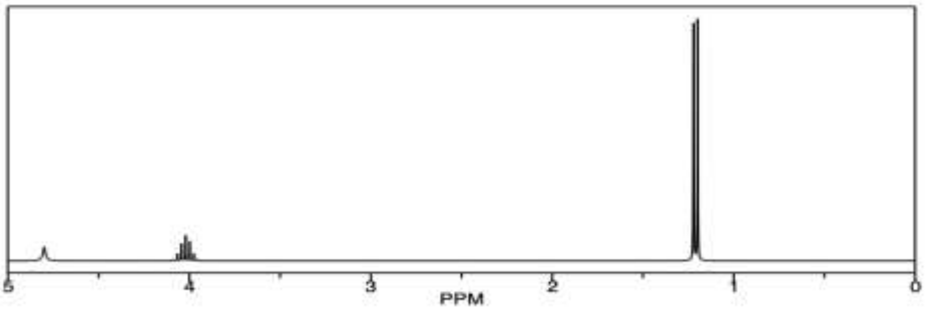
Product B

^{13}C-NMR



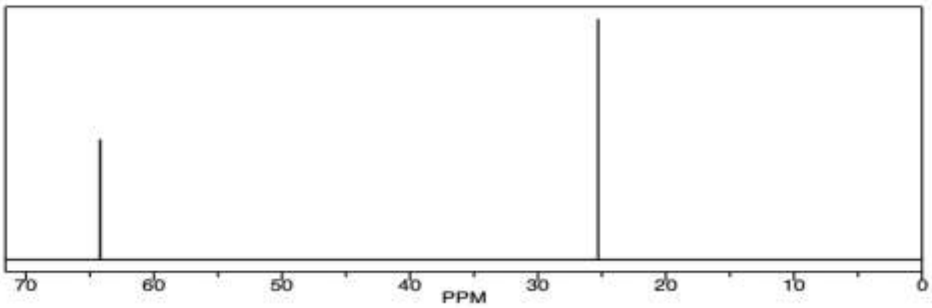
Product C

^{1}H-NMR



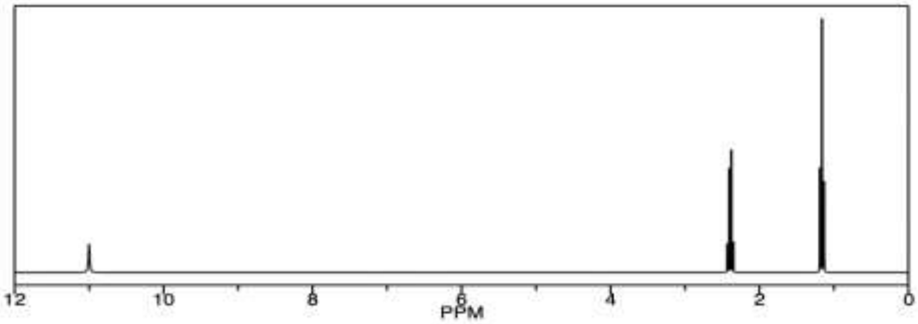
Product C

^{13}C-NMR



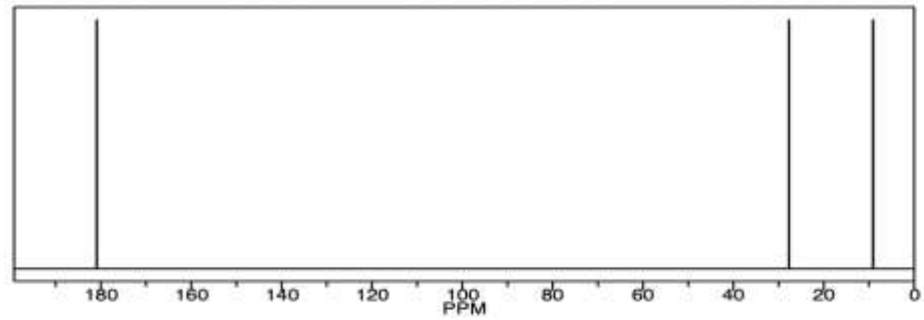
Compound D

^{1}H-NMR



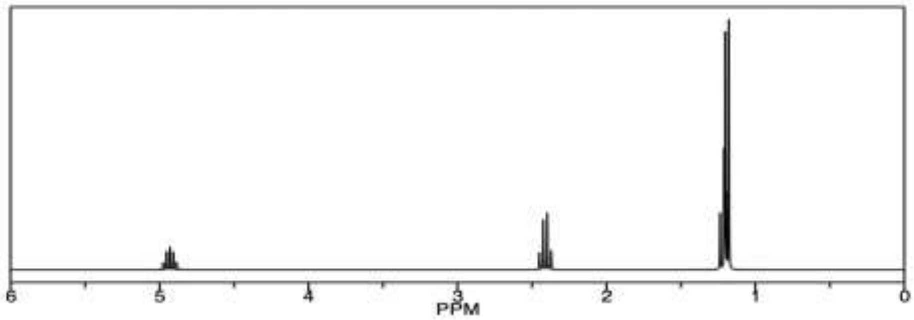
Compound D

^{13}C-NMR



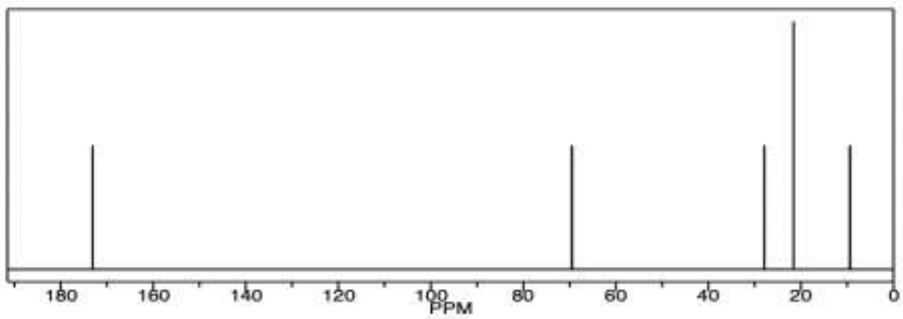
Compound E

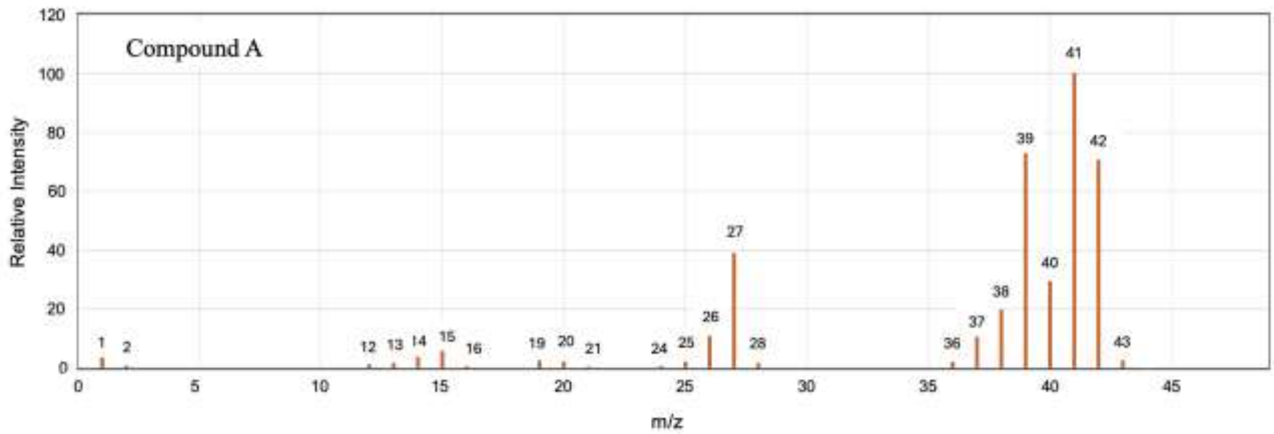
^{1}H-NMR



Compound E

^{13}C-NMR





^{1}H NMR chemical shift data

Type of proton		δ/ppm
$Si(CH_3)_4$	(TMS)	0
$R—CH_3$		0.7–1.3
$R—CH_2—R$		1.2–1.5
$R—CHR_2$		1.5–2.0
$H_3C—CO—$	(aldehydes, ketones or esters)	2.0–2.5
$—CH—CO—$	(aldehydes, ketones or esters)	2.1–2.6
$H_3C—O—$	(alcohols or esters)	3.2–4.0
$—CH—O—$	(alcohols or esters)	3.3–5.1
$R_2—CH_2—O—$	(alcohols or esters)	3.5–5.0
$R—OH$		1–6
$R_2C{=}CHR$	(alkene)	4.5–7.0
$R—CHO$	(aldehyde)	9.4–10.0
$R—COOH$		9.0–13.0

(a) Determine the structures and names of compounds A – E. Justify your responses (overpage) with reference to specific aspects of the data provided. *(8 marks)*

Structure and Name	**Structure and Name**
A:	**B**: ....
C:	**D**: ....
E:	

Justification

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(b) Predict one difference you would expect to see between the IR spectra of D and E.

(1 mark)

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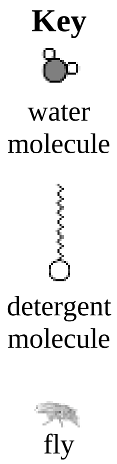
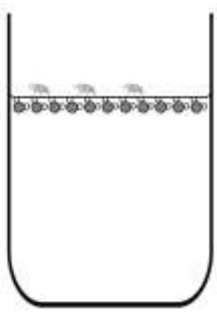
Question 31. (5 marks)

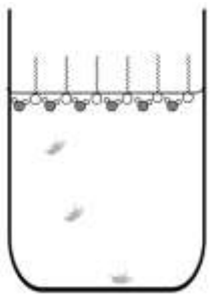
Surface tension of water results from hydrogen bonding between water molecules, allowing small insects to land on the surface of water without falling in. Detergent reduces the surface tension of water by disrupting the hydrogen bonding. Gardeners take advantage of this phenomenon to make an insect trap with apple cider vinegar, water and a few drops of detergent.

Flies attracted to the trap drown as they cannot land on the surface. The following diagrams illustrate the effect that detergent molecules have on the surface tension of water.





Water molecules are attracted to each other and create surface tension.
(Molecules and flies are not to scale!)



Detergent molecules disrupt the forces of attraction between water molecules, reducing surface tension.

(a) Explain why the diagram shows the detergent molecules with one end sticking out from the surface of the water. *(2 marks)*

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(b) Using a similar symbol as above for detergent molecules, draw labelled diagrams to illustrate how detergent lifts grease from dirty clothing. *(3 marks)*

Question 32. (8 marks)

The solubility of an ionic salt is a function of how easily water molecules break the ionic bonds in the crystal lattice of the ionic solid. The energy required to break the ionic bonds is provided by the energy given out as ion-dipole bonds form between the water molecules and the ions. The difference between these two energies determines the overall enthalpy change for the solvation (hydration) process (ΔH_{hydr}) and can be either exothermic or endothermic.

The following table lists the molar solubilities of the carbonates of three Group II metals and the relative enthalpies of hydration for the metal cations.

	$MgCO_3$	**$CaCO_3$**	**$BaCO_3$**
Solubility ($mol.L^{-1}$)	2.61×10^{-3}	5.8×10^{-5}	5.08×10^{-5}
ΔH_{hydr} ($kJ.mol^{-1}$)	-1921	-1577	-1305
Atomic number of metal cation	12	20	56

(a) What can you conclude from the data above about the relative ease with which the Group II cations form ion-dipole bonds with water as you descend a group in the Periodic Table? Explain your reasoning. *(2 marks)*

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Question continues over page.

(b) $Ca(OH)_2$ has a K_{sp} of 5.02×10^{-6} , which is very similar to the K_{sp} value of $MgCO_3$ (6.82×10^{-6}). Based on this observation, a student stated that the molar solubilities of $Ca(OH)_2$ and $MgCO_3$ would therefore be very similar.

Assess the validity of the student's statement. Support your assessment using relevant equations and calculations. *(4 marks)*

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(c) If 50 mL of a 0.050 M $Ca(NO_3)_2$ solution is mixed with 50 mL of a 0.050 M NaOH solution, will a precipitate form? Show your working. *(2 marks)*

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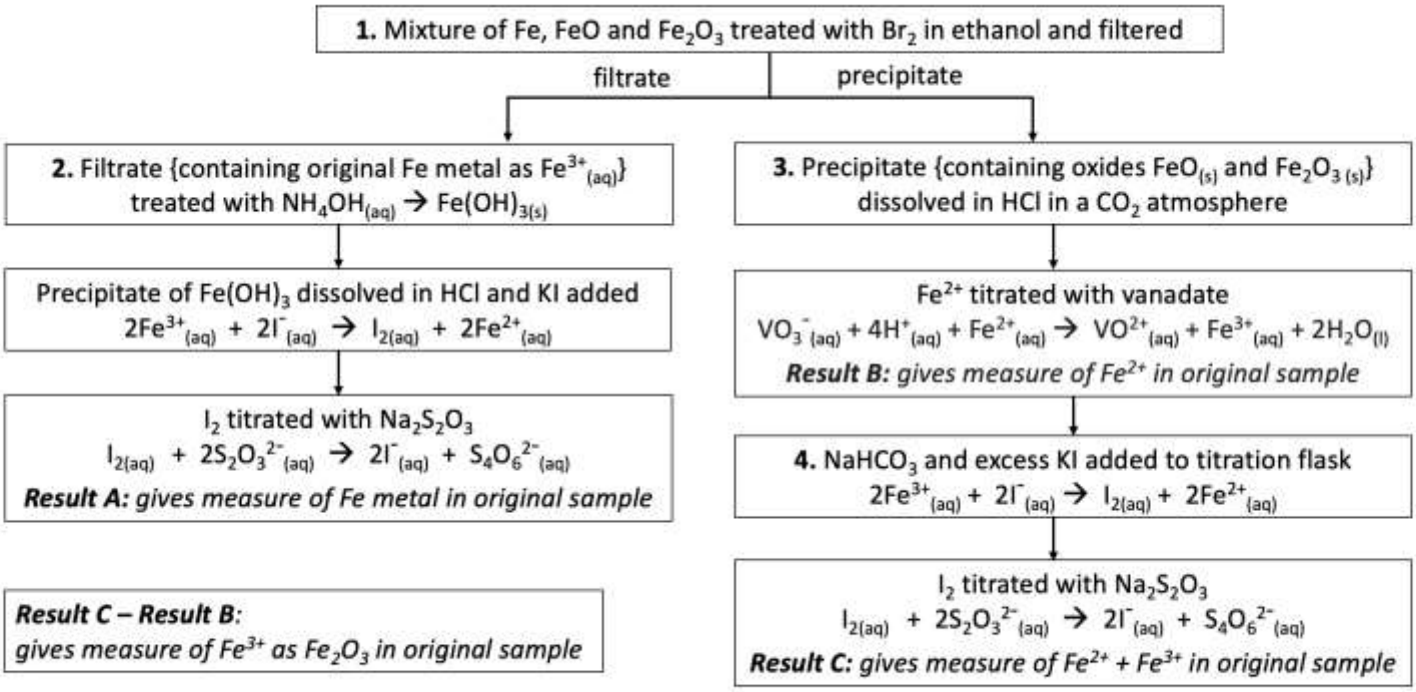
Question 33. (6 marks)

An impure mixture of metallic iron and iron oxides was analysed to determine the total percentage iron content by the following procedure.[4]

1. A 150 mg sample of the mixture was treated with bromine in ethanol to dissolve the metallic iron, forming iron (III) bromide, followed by filtration.
2. The filtrate was treated with NH_4OH to precipitate the metallic iron as $Fe(OH)_3$, which was dissolved in HCl and treated with KI solution to form elemental iodine. The iodine was titrated with sodium thiosulphate, using a starch indicator.
   5.3 mL of 0.20 M $Na_2S_2O_3$ was required to fully react with the Fe^{3+}.
3. The original precipitate, containing the iron oxides (FeO and Fe_2O_3), was dissolved in HCl in a CO_2 atmosphere, then titrated with vanadate to determine the amount of Fe^{2+} present.
   5.8 mL of 0.10 M $NaVO_3$ was required to fully oxidise the Fe^{2+} to Fe^{3+}.
4. The amount of Fe^{3+} present after step 3 was determined by reaction with $NaHCO_3$ and KI to form elemental iodine, followed by titration with sodium thiosulphate, using a starch indicator.
   6.9 mL of 0.20 M $Na_2S_2O_3$ was required to fully react with the Fe^{3+}.

The procedure and relevant equations are summarised in the flowchart below.



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4 Sant, BR and Prasad, TP (1968) Determination of metallic iron, iron(II) oxide, and iron(III) oxide in a mixture. *Talanta 15(12)*: 1483-1486

(a) Determine the mass of iron present in the original sample as metallic iron. *(2 marks)*

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(b) Determine the mass of iron present in the original sample as Fe^{2+}. *(1 mark)*

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(c) Determine the mass of iron present in the original sample as Fe^{3+}. *(2 marks)*

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(d) Determine the percentage by mass of total iron in the original sample. *(1 mark)*

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Question 34. (4 marks)

The following table lists the boiling points of three homologous series of straight-chained organic compounds (A, B and C) but the column headings are missing. The three series of compounds, in alphabetical order, are alcohols, aldehydes and alkanes.

	Boiling Point (°C)		
No. of carbon atoms	A ____________________	B ____________________	C ____________________
1	- 162	65	- 21
2	- 89	78	21
3	- 42	97	49
4	- 1	117	76
5	36	138	102
6	69	158	131
7	98	176	178
8	126	195	193

(a) Add the relevant column headings to the table. *(1 mark)*

(b) Justify your assignment of column headings, with reference to the structures of the three types of functional group and the resultant intermolecular forces between molecules. *(2 marks)*

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(c) Explain why the boiling points in any homologous series increase with increasing molar mass. *(1 mark)*

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Question 35. (4 marks)

Consider the following equilibrium reaction.

$$SO_{2(g)} + Cl_{2(g)} \rightleftharpoons SO_2Cl_{2(g)}$$

4.902×10^{-2} moles of $SO_{2(g)}$ and 4.902×10^{-2} moles of $Cl_{2(g)}$ were introduced into a 5 L container at 375°C. After the reaction reached equilibrium, the concentration of $SO_2Cl_{2(g)}$ was 9.75×10^{-4} M.

(a) Write the expression for the equilibrium constant for this reaction. *(1 mark)*

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(b) Determine the value of the equilibrium constant at 375°C. *(3 marks)*

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THIS IS THE END OF THE EXAMINATION PAPER.

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

PEM

2025

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Chemistry

Solutions, Mapping Grid and Marking Guidelines

Section I: Multiple Choice Solutions and Mapping Grid

1 mark each

Qu	Answer	Syllabus Section	Outcomes
1	A	Mod 5 Calculating the Equilibrium Constant (K_{eq})	CH12-6, CH12-12
2	D	Mod 5 Factors that Affect Equilibrium	CH12-6, CH12-12
3	A	Mod 5 Calculating the Equilibrium Constant (K_{eq})	CH12-12
4	B	Mod 5 Solution Equilibria	CH12-6, CH12-12
5	C	Mod 5 Static and Dynamic Equilibrium Mod 6 Properties of Acids and Bases	CH12-12, CH12-13
6	D	Mod 6 Properties of Acids and Bases	CH12-13
7	C	Mod 6 Using Brønsted-Lowry Theory	CH12-13
8	D	Mod 6 Quantitative Analysis	CH12-4, CH12-13
9	A	Mod 6 Properties of Acids and Bases	CH12-6, CH12-13
10	B	Mod 7 Reactions of Organic Acids and Bases	CH12-14
11	D	Mod 8 Analysis of Inorganic Substances	CH12-15
12	B	Mod 6 Quantitative Analysis	CH12-5, CH12-13
13	C	Mod 7 Reactions of Organic Acids and Bases	CH12-14
14	A	Mod 7 Nomenclature Mod 7 Polymers	CH12-5, CH12-14
15	B	Mod 7 Polymers	CH12-6, CH12-14
16	A	Mod 7 Alcohols	CH12-14
17	D	Mod 8 Analysis of Inorganic Substances	CH12-15
18	B	Mod 8 Analysis of Organic Substances	CH12-6, CH12-15
19	C	Mod 8 Analysis of Organic Substances	CH12-5, CH12-14
20	C	Mod 8 Analysis of Inorganic Substances	CH12-6, CH12-15

Section II: Short-answer Questions Mapping Grid

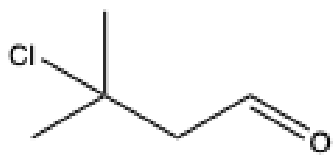
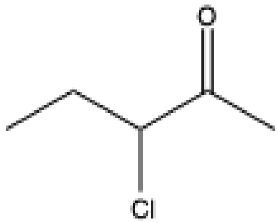
Qu	Marks	Syllabus Section	Syllabus Outcomes
21	2	Mod 7 Nomenclature	CH12-14
22	3	Mod 6 Quantitative Analysis	CH12-7, CH12-13
23	4	Mod 6 Using Brønsted-Lowry Theory	CH12-7, CH12-13
24a	1	Mod 6 Using Brønsted-Lowry Theory	CH12-6, CH12-13
24b	3	Mod 6 Quantitative Analysis	CH12-6, CH12-13
25a	1	Mod 6 Properties of Acids and Bases	CH12-6, CH12-13
25bi	1	Mod 5 Calculating the Equilibrium Constant (K_{eq})	CH12-6, CH12-12
25bii	2	Mod 6 Quantitative Analysis	CH12-6, CH12-12
25c	2	Mod 6 Quantitative Analysis	CH12-6, CH12-7, CH12-13
25d	1	Mod 8 Analysis of Inorganic Substances	CH12-6, CH12-15
25e	2	Mod 8 Analysis of Inorganic Substances	CH12-4, CH12-7, CH12-15
26a	1	Mod 6 Properties of Acids and Bases	CH12-13
26b	3	Mod 6 Properties of Acids and Bases	CH12-6, CH12-13
26c	2	Mod 6 Quantitative Analysis	CH12-6, CH12-13
27a	2	Mod 5 Factors that Affect Equilibrium	CH12-7, CH12-12
27b	2	Mod 5 Factors that Affect Equilibrium	CH12-7, CH12-12
28a	1	Mod 8 Chemical Synthesis and Design	CH12-15
28b	2	Mod 8 Chemical Synthesis and Design	CH12-15
28c	2	Mod 7 Polymers	CH12-6, CH12-14
28d	4	Mod 7 Hydrocarbons	CH12-7, CH12-14
29b	3	Mod 8 Analysis of Inorganic Substances	CH12-15
30a	8	Mod 7 Nomenclature Mod 7 Products of Reactions Involving Hydrocarbons Mod 8 Analysis of Organic Substances	CH12-5, CH12-6, CH12-7, CH12-14, CH12-15
30b	1	Mod 8 Analysis of Organic Substances	CH12-7, CH12-15
31a	2	Mod 7 Reactions of Organic Acids and Bases	CH12-7, CH12-14
31b	3	Mod 7 Reactions of Organic Acids and Bases	CH12-14
32a	2	Mod 5 Solution Equilibria	CH12-5, CH12-12
32b	4	Mod 5 Solution Equilibria	CH12-5, CH12-7, CH12-12
32c	2	Mod 5 Solution Equilibria	CH12-6, CH12-12
33a	2	Mod 8 Analysis of Inorganic Substances	CH12-4, CH12-6, CH12-15
33b	1	Mod 8 Analysis of Inorganic Substances	CH12-4, CH12-6, CH12-15
33c	2	Mod 8 Analysis of Inorganic Substances	CH12-4, CH12-6, CH12-15
33d	1	Mod 8 Analysis of Inorganic Substances	CH12-4, CH12-6, CH12-15
34a	1	Mod 7 Hydrocarbons	CH12-14
34b	2	Mod 7 Alcohols	CH12-7, CH12-14
34b	1	Mod 7 Alcohols	CH12-7, CH12-14
35a	1	Mod 5 Calculating the Equilibrium Constant (Keq)	CH12-12
35b	3	Mod 5 Calculating the Equilibrium Constant (Keq)	CH12-6, CH12-12

Section II Marking Guidelines

Question 21

Criteria	Marks
• Draws and names one chain isomer and one functional group isomer of 3-chloropentanal	**2**
• Draws and names one chain isomer or one functional group isomer of 3-chloropentanal OR • Draws one chain isomer and one functional group isomer of 3-chloropentanal	**1**

Sample Answer

Chain isomer	Functional group isomer
	
Name: 3-chloro-3-methylbutanal	Name: 3-chloropentan-2-one

Question 22

Criteria	Marks
• Chooses two of the compounds that make up a conjugate acid-base pair • Explains how a solution of a weak acid and its conjugate base (or weak base and its conjugate acid) behaves as a buffer • Includes a relevant chemical equation	**3**
• Provides two of the above	**2**
• Provides one of the above	**1**

Sample Answer

Solutions of sodium dihydrogen phosphate and sodium hydrogen phosphate mixed in equal proportions will behave as a buffer. $H_2PO_4^-$ is a weak acid and HPO_4^{2-} is its conjugate base. A buffer acts to resist changes in pH.

$$H_2PO_4^-{}_{(aq)} \rightleftharpoons HPO_4^{2-}{}_{(aq)} + H^+{}_{(aq)}$$

If acid is added to the buffer mixture, the HPO_4^{2-} will take up the excess H^+ ions to form more $H_2PO_4^-$ and the pH will not change. If a base is added to the buffer mixture, the added OH^- ions will react with H^+ ions, the equilibrium will shift to the right to form more H^+ ions and the pH will not change.

Question 23

Criteria	Marks
• Defines a Brønsted-Lowry acid • Provides an equation showing water acting as a proton donor • Defines a Brønsted-Lowry base • Provides an equation showing water acting as a proton acceptor	**4**
• Provides three of the above	**3**
• Provides two of the above	**2**
• Provides one of the above	**1**

Sample Answer

A Brønsted-Lowry acid is defined as a substance that acts as a proton donor. Water acts as a proton donor in the presence of a base.

$$H_2O_{(l)} + NH_{3(aq)} \rightleftharpoons NH_4^+{}_{(aq)} + OH^-{}_{(aq)}$$

A Brønsted-Lowry base is defined as a proton acceptor. Water acts as a proton acceptor in the presence of an acid.

$$H_2O_{(l)} + CH_3COOH_{(aq)} \rightleftharpoons CH_3COO^-{}_{(aq)} + H_3O^+{}_{(aq)}$$

Question 24(a)

Criteria	Marks
• Calculates the pH of 0.5% KOH	**1**

Sample Answer

0.5 g/100 mL of KOH = (0.5 g/56.108 g.mol^{-1}) / 100 mL = 0.0089 mol / 100 mL = 0.089 mol/L

pOH = $-\log_{10}(0.089)$ = 1.05

pH = 12.95

Question 24(b)

Criteria	Marks
• Calculates the molarity and percent concentration of the KOH	**3**
• Provides some steps in the calculation	**2**
• Provides some relevant information	**1**

Sample Answer

average volume of 0.250 M HCl = 16.2 mL

no. mol HCl = 0.250 mol.L^{-1} x 0.0162 L = 0.00405 mol

$KOH_{(aq)} + HCl_{(aq)} \rightarrow KCl_{(aq)} + H_2O_{(l)}$

no. mol KOH = 0.00405 mol

[KOH] = 0.00405 mol / 0.050 L = 0.0810 M

% m/v = (0.0810 mol.L^{-1} x 56.108 g.mol^{-1}) ÷ 10 = 0.454 g / 100 mL = 0.454%

Question 25(a)

Criteria	Marks
• Calculates the molarity of 0.2% methyl red solution	**1**

Sample Answer

(1.00 g / 269.3 g.mol^{-1}) / 0.500 L = 0.0074 M

Question 25(b)(i)

Criteria	Marks
• Determines the pH of a 0.2% methyl red solution that is 50% ionised	1

Sample Answer

When equal amounts of the acid and base are present, the pH = pK_a = 4.95

Question 25(b)(ii)

Criteria	Marks
• Determines the pH of a 0.2% methyl red solution that is 90% ionised	2
• Provides some relevant information	1

Sample Answer

$pK_a = 4.95$

$K_a = 10^{-4.95} = 1.12 \times 10^{-5}$

If 90% ionised: $[MRH] = 0.0074 \times 0.1 = 0.00074$ M

and $[MR^-] = 0.0074 \times 0.9 = 0.0067$ M

$$K_a = \frac{[MR^-][H^+]}{[MRH]}$$

$$[H^+] = \frac{K_a[MRH]}{[MR^-]} = \frac{1.12 \times 10^{-5} \times 0.00074}{0.0067}$$

$$= 1.24 \times 10^{-6} \text{ M}$$

$pH = -\log_{10}(1.24 \times 10^{-6}) = 5.9$

Students who recall the Henderson-Hasselbach equation may calculate it as follows:

$$pH = pK_a + \log_{10}\frac{[A^-]}{[HA]}$$

$$pH = 4.95 + \log_{10}\frac{[MR^-]}{[MRH]}$$

$$pH = 4.95 + \log_{10}(0.0067/0.00074)$$

$$pH = 5.9$$

Question 25(c)

Criteria	Marks
• Indicates the suitability of methyl red as an indicator for a strong base/weak acid titration and provides an explanation	2
• Provides some relevant information	1

Sample Answer

The pK_a of 4.95 indicates that methyl red changes colour at pH < 7. The pH at the endpoint of a titration between a strong base and a weak acid is in the basic range so methyl red is not a suitable indicator for the titration of NaOH and acetic acid.

Question 25(d)

Criteria	Marks
• Calculates the molarity of methyl red solution using the Beer-Lambert equation	**1**

Sample Answer

Rearranging the Beer-Lambert equation:

$c = A / \varepsilon \ell = 0.315 / (2.30 \times 10^4 \text{ L.mol}^{-1}.\text{cm}^{-1} \times 0.200 \text{ cm}) = 6.85 \times 10^{-5} \text{ M}$

Question 25(e)

Criteria	Marks
• Determines the best wavelength to use for a colourimetric analysis of methyl red in basic solution and provides an explanation	**2**
• Provides some relevant information	**1**

Sample Answer

The wavelength used in colourimetry should be the wavelength at which the absorption of light by the compound to be measured is at a maximum. According to the graph on the right, this is approximately 430 nm at pH 12 for methyl red.

Question 26(a)

Criteria	Marks
• Provides a balanced equation for the reaction between $NaHCO_3$ and HCl, including correct subscripts	**1**

Sample Answer

$NaHCO_{3(aq)} + HCl_{(aq)} \rightarrow H_2O_{(l)} + NaCl_{(aq)} + CO_{2(g)}$

Question 26(b)

Criteria	Marks
• Calculates the molar enthalpy of neutralisation in $kJ.mol^{-1}$	**3**
• Provides some steps in the calculation	**2**
• Provides some relevant information	**1**

Sample Answer

heat evolved by neutralisation = $0.150 \text{ kg} \times 4.18 \times 10^3 \text{ J.kg}^{-1}.\text{K}^{-1} \times 7 \text{ K} = 4389 \text{ J} = 4.389 \text{ kJ}$

moles of HCl = moles of $NaHCO_3$ = $0.075 \text{ L} \times 1 \text{ mol.L}^{-1} = 0.075 \text{ mol}$

molar enthalpy of neutralisation = $4.389 \text{ kJ} / 0.075 \text{ mol} = 58.5 \text{ kJ.mol}^{-1}$

Question 26(c)

Criteria	Marks
• Calculates the volume of CO_2 evolved in the reaction between $NaHCO_3$ and HCl	**2**
• Provides some relevant information	**1**

Sample Answer

moles of CO_2 evolved = 0.075 mol

volume of CO_2 at 25°C = $0.075 \text{ mol} \times 24.79 \text{ L.mol}^{-1} = 1.86 \text{ L}$

Question 27(a)

Criteria	Marks
• Predicts and explains the effect that adding solid NH_4HS has on the equilibrium partial pressure of NH_3	**2**
• Provides some relevant information	**1**

Sample Answer

The partial pressure of NH_3 will not change. Adding more solid does not increase the concentration of the solid so no net reaction will occur.

Question 27(b)

Criteria	Marks
• Predicts and explains the effect of increasing the temperature on the amount of $NH_4HS_{(s)}$ present	**2**
• Provides some relevant information	**1**

Sample Answer

The amount of solid NH_4HS will decrease. The forward reaction is endothermic so if the temperature is increased, the forward reaction, which absorbs heat, will be favoured.

Question 28(a)

Criteria	Marks
• Outlines the role of acid-catalysed hydrolysis in the REACH process	**1**

Sample Answer

Hydrolysis refers to the breaking of bonds with the addition of water. Acid acts as a catalyst to increase the rate of this reaction. (The bonds broken are those between the glucose molecules in the carbohydrates in the bagasse.)

Question 28(b)

Criteria	Marks
• Provides two factors that would need to be considered when deciding on the location of a plant for producing aviation fuel from bagasse	**2**
• Provides some relevant information	**1**

Sample Answer

The plant should be located in close proximity to sugarcane farms to provide the raw material (bagasse). It should also be close to transport, such as a railway or shipping port, for efficient distribution of the fuel to airports where it is used.

Question 28(c)

Criteria	Marks
• Provides an explanation for the superior strength and thermal stability of PEF compared to PET, with reference to the difference in the structure of the two molecules	**2**
• Provides some relevant information	**1**

Sample Answer

PEF is a linear molecule and PET is bent. Linear molecules can stack together more closely than bent molecules, increasing the potential for dispersion forces between adjacent molecules to hold them together. This means more force is required to separate the strands, making PEF a stronger and more thermally-stable product than PET.

Question 28(d)

Criteria	Marks
• Suggests at least four economic and environmental advantages of using bagasse in the REACH process to make aviation fuel and plastics	**4**
• Suggests three economic and/or environmental advantages of using bagasse in the REACH process to make aviation fuel and plastics	**3**
• Suggests two economic and/or environmental advantages of using bagasse in the REACH process to make aviation fuel and plastics	**2**
• Provides some relevant information	**1**

Sample Answer

Any four reasonable suggestions, e.g.

- Producing plastic materials and fuels from a by-product of sugarcane farming reduces the environmental impact of waste.
- It is more economical to use a waste product than it is to mine for petroleum.
- The REACH process is sustainable, as the raw material is acquired from a crop plant. Petroleum that has traditionally been used to produce fuel and plastics is a non-renewable resource.
- Converting bagasse to useful products such as plates and cutlery instead of burning it reduces carbon dioxide pollution.
- Returning the unreacted fibrous material to the soil creates a carbon sink, reducing carbon dioxide pollution.
- Ethanol that is produced as a by-product is recycled back into the manufacturing process.

Question 29

Criteria	Marks
• Outlines a sequence of suitable precipitation tests with expected observations to identify the three anions • Includes balanced chemical equations, with correct subscripts, for all positive tests	**3**
• Outlines one suitable precipitation test with expected observations • Includes a balanced chemical equation, with correct subscripts, for the test	**2**
• Provides some relevant information	**1**

Sample Answer

Add several drops of barium nitrate solution to a sample of the unknown solution. A white precipitate indicates that the anion is sulphate. $Ba^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}$

Add several drops of silver nitrate solution to a sample of the unknown solution. A white precipitate indicates that the anion is iodide. $Ag^{+}_{(aq)} + I^{-}_{(aq)} \rightarrow AgI_{(s)}$

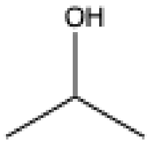
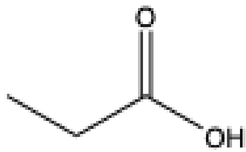
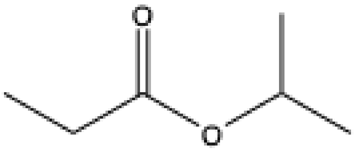
(Or add several drops of lead (II) nitrate solution to a sample of the solution. A yellow precipitate indicates the anion is iodide. $Pb^{2+}_{(aq)} + 2I^{-}_{(aq)} \rightarrow PbI_{2(s)}$)

If the unknown solution gives a negative result for both tests, it is sodium acetate.

Question 30(a)

Criteria	Marks
• Provides the correct structure and name for compounds A, B, C, D and E • Demonstrates an extensive understanding of the interpretation of spectroscopic data • Demonstrates an extensive understanding of the addition reactions of unsaturated hydrocarbons and the esterification reaction between an acid and an alcohol • Justifies the identification of each compound with specific reference to relevant data from the spectra and reactions provided	**8**
• Provides the correct structure and name for four compounds • Demonstrates a thorough understanding of the interpretation of spectroscopic data • Demonstrates a thorough understanding of the addition reactions of unsaturated hydrocarbons and the esterification reaction between an acid and an alcohol • Justifies the identification of each compound with specific reference to relevant data from the spectra and reactions provided	**6-7**
• Provides the correct structure and name for three compounds • Demonstrates some understanding of the interpretation of spectroscopic data • Demonstrates some understanding of the addition reactions of unsaturated hydrocarbons and/or the esterification reaction between an acid and an alcohol • Justifies the identification of each compound with specific reference to relevant data from the spectra or reactions provided	**4-5**
• Provides the correct structure and name for two compounds • Demonstrates some understanding of the interpretation of spectroscopic data • Justifies the identification of each compound with specific reference to relevant data from the spectra or reactions provided	**2-3**
• Provides some relevant information	**1**

Sample Answer

Structure and Name	Structure and Name
A:  propene	**B**:  propan-1-ol
C:  propan-2-ol	**D**:  propanoic acid
E:  isopropyl propanoate	(or isopropyl propionate/propan-2-yl propanoate)

Justification

The M+1 signal on the mass spectrum of Compound A is 43 so the molar mass is 42. Compound A must have the formula C_3H_6 so is propene.

Compounds B and C are the major and minor products of the addition of water to propene so are both alcohols. This is supported by the fact that each has a singlet at 4.7-4.8 ppm on the ^{1}H-NMR, which is characteristic of an alcohol H adjacent to a -CH- or a $-CH_2-$ group.

B must be propan-1-ol, as it has three types of C atoms and four types of H atoms. The sextet at 1.6 ppm is from the hydrogens on the second carbon, with five neighbours.

C must be propan-2-ol, as it only has two types of C atoms and three types of H atoms. The six methyl hydrogens have only one neighbouring H, giving the large doublet at 1.2 ppm.

D is the product of oxidation of B (propan-1-ol) so must be propanoic acid. This is supported by the presence of a signal at 180 ppm on the ^{13}C-NMR, characteristic of the carbonyl group of a carboxylic acid, and 11 ppm on the ^{1}H-NMR, characteristic of the -COOH hydrogen of a carboxylic acid.

When compound D was reacted under suitable conditions with the second product of the initial reaction (C), the resultant product (E) had to be isopropyl propanoate. This is supported by the presence of five signals on the ^{13}C-NMR but only three signals on the ^{1}H-NMR of compound E. (The large signal at 1.2 ppm on the ^{1}H-NMR is due to overlapping signals from the nine methyl hydrogens.)

Question 30(b)

Criteria	Marks
• Correctly predicts one difference you would expect to see between the IR spectra of D and E	**1**

Sample Answer

Compound D is a carboxylic acid so you would expect to see a broad band at 2500 - 3000 cm^{-1} in the IR spectrum, characteristic of the -OH group of a carboxylic acid. The ester, E, has no -OH group so its IR spectrum would not contain this band.

Question 31(a)

Criteria	Marks
• Provides a reason why the detergent molecules are shown with one end sticking out from the surface of the water	2
• Provides some relevant information	1

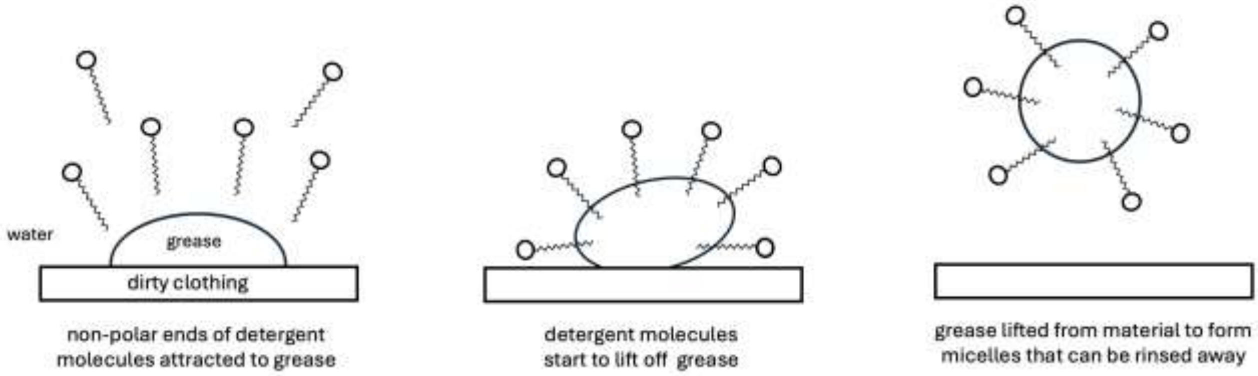
Sample Answer

One end of the detergent molecule is polar so is attracted to the polar water molecules. The other end is non-polar and is repelled by the water molecules, so is shown sticking out from the surface of the water.

Question 31(b)

Criteria	Marks
• Provides appropriate, labelled diagrams to demonstrate a clear understanding of how detergent lifts grease from dirty clothing	3
• Demonstrates some understanding of how detergent lifts grease from dirty clothing	2
• Provides some relevant information	1

Sample Answer



Question 32(a)

Criteria	Marks
• Relates the trend in the solubility of the carbonates and their ΔH_{hydr} value to the ease with which the cation forms ion-dipole bonds with water	2
• Provides some relevant information	1

Sample Answer

The ease with which the Group II cations form ion-dipole bonds with water decreases down the group. The smallest cation, Mg^{2+}, forms these bonds the most easily as it has the highest ΔH_{hydr} and its carbonate is the most soluble in water. The energy released when the ions become hydrated decreases down the group and this correlates with decreasing solubility.

Question 32(b)

Criteria	Marks
• Assesses the validity of the statement that the molar solubilities of $Ca(OH)_2$ and $MgCO_3$ would be very similar • Supports the assessment with relevant equations and calculation of the molar solubility of $Ca(OH)_2$	**4**
• Assesses the validity of the statement that the molar solubilities of $Ca(OH)_2$ and $MgCO_3$ would be very similar • Supports the assessment with some steps in the calculation of the molar solubility of $Ca(OH)_2$	**3**
• Provides some steps in the calculation	**2**
• Provides some relevant information	**1**

Sample Answer

The student's statement is invalid. The molar solubilities of $Ca(OH)_2$ and $MgCO_3$ cannot be assumed to be similar based on the similarity of their K_{sp} values.

For $Ca(OH)_2$, the solution equation is: $Ca(OH)_{2(s)} \rightleftharpoons Ca^{2+}_{(aq)} + 2OH^-_{(aq)}$

$K_{sp} = [Ca^{2+}][OH^-]^2$

Let the molar solubility = x

In a saturated solution, $[Ca^{2+}] = x$ mol.L^{-1} and $[OH^-] = 2x$ mol.L^{-1}

$K_{sp} = (2x)^2 \times (x) = 4x^3$

$x = \sqrt[3]{5.02 \times 10^{-6} / 4} = 0.0108$ mol.L^{-1}

For $MgCO_3$, the solution equation is: $MgCO_{3(s)} \rightleftharpoons Mg^{2+}_{(aq)} + CO_3{}^{2-}_{(aq)}$

$K_{sp} = [Mg^{2+}][CO_3^{2-}]$

Let the molar solubility = x

In a saturated solution, $[Mg^{2+}] = x$ mol.L^{-1} and $[CO_3^{2-}] = x$ mol.L^{-1} so $K_{sp} = x^2$

$x = \sqrt{6.82 \times 10^{-6}} = 0.00261$ mol.L^{-1}

(Students may, of course, simply cite this value from the table above and don't need to calculate it!)

The molar solubility of $Ca(OH)_2$ is approximately 4 times that of $MgCO_3$.

Question 32(c)

Criteria	Marks
• Determines if a precipitate will form by comparing the ionic product with the K_{sp} value of $Ca(OH)_2$	**2**
• Provides some relevant information	**1**

Sample Answer

K_{sp} of $Ca(OH)_2 = [Ca^{2+}][OH^-]^2 = 5.02 \times 10^{-6}$

$[Ca^{2+}] = (0.050 \text{ mol.L}^{-1} \times 0.050 \text{ L}) / 0.100 \text{ L} = 0.025$ M

$[OH^-] = (0.050 \text{ mol.L}^{-1} \times 0.050 \text{ L}) / 0.100 \text{ L} = 0.025$ M

Ionic product = $(0.025)^3 = 0.0000156 = 1.56 \times 10^{-5}$

IP > K_{sp} so a precipitate will form.

Question 33(a)

Criteria	Marks
• Determines the mass of iron present in the original sample as metallic iron	2
• Provides some relevant information	1

Sample Answer

No. moles of $S_2O_3^{2-}$ used in titration = 0.0053 L x 0.2 mol.L^{-1} = 1.06 x 10^{-3} mol

Titration equation: $I_{2(aq)} + 2S_2O_3^{2-}{}_{(aq)} \rightarrow 2I^-{}_{(aq)} + S_4O_6^{2-}{}_{(aq)}$

So, no. moles I_2 = (no. moles $S_2O_3^{2-}$) / 2 = 5.3 x 10^{-4} mol

Reaction to produce I_2: $2Fe^{3+}{}_{(aq)} + 2I^-{}_{(aq)} \rightarrow I_{2(aq)} + 2Fe^{2+}{}_{(aq)}$

No. moles Fe^{3+} = 5.3 x 10^{-4} mol x 2 = 1.06 x 10^{-3} mol

Mass Fe^{3+} from metallic iron = 1.06 x 10^{-3} mol x 55.85 g.mol^{-1} = 0.05920 g

Question 33(b)

Criteria	Marks
• Determines the mass of iron present in the original sample as Fe^{2+}	1

Sample Answer

No. moles of VO_3^- used in titration = 0.0058 L x 0.1 mol.L^{-1} = 5.8 x 10^{-4} mol

Titration equation: $VO_3^-{}_{(aq)} + 4H^+{}_{(aq)} + Fe^{2+}{}_{(aq)} \rightarrow VO^{2+}{}_{(aq)} + Fe^{3+}{}_{(aq)} + 2H_2O_{(l)}$

So, no. moles Fe^{2+} = 5.8 x 10^{-4} mol

Mass of iron present in original sample as Fe^{2+} = 5.8 x 10^{-4} mol x 55.85 g.mol^{-1} = 0.03239 g

Question 33(c)

Criteria	Marks
• Correctly determines the mass of iron present in the original sample as Fe^{3+}	2
• Provides some relevant information	1

Sample Answer

No. moles of $S_2O_3^{2-}$ used in titration = 0.0069 L x 0.2 mol.L^{-1} = 1.38 x 10^{-3} mol

Titration equation: $I_{2(aq)} + 2S_2O_3^{2-}{}_{(aq)} \rightarrow 2I^-{}_{(aq)} + S_4O_6^{2-}{}_{(aq)}$

So, no. moles I_2 = (no. moles $S_2O_3^{2-}$) / 2 = 6.9 x 10^{-4} mol

Reaction to produce I_2: $2Fe^{3+}{}_{(aq)} + 2I^-{}_{(aq)} \rightarrow I_{2(aq)} + 2Fe^{2+}{}_{(aq)}$

No. moles Fe^{3+} = 6.9 x 10^{-4} mol x 2 = 1.38 x 10^{-3} mol

Mass of Fe^{3+} (+ original Fe^{2+}) = 1.38 x 10^{-3} mol x 55.85 g.mol^{-1} = 0.07707 g

Mass Fe^{3+} present in the original sample as Fe^{3+} = 0.07707 - 0.03239 g = 0.04468 g

Question 33(d)

Criteria	Marks
• Correctly determines the percentage by mass of total iron in the original sample	1

Sample Answer

% by mass of total Fe in original sample = [(0.05920 + 0.07707) / 0.150] x 100 = 91%

Question 34(a)

Criteria	Marks
• Correctly identifies the three column headings	1

Sample Answer

A – alkanes ; B – alcohols ; C - aldehydes

Question 34(b)

Criteria	Marks
• Justifies the assignment of column headings, with reference to structure and intermolecular forces between molecules	2
• Provides some relevant information	1

Sample Answer

The alkanes have the lowest boiling points as they are non-polar molecules and are attracted to each other only by weak dispersion forces. The boiling points of the aldehydes are the next highest, as the C=O bond is polar and allows for dipole-dipole attractions between molecules. Alcohols have the highest boiling points for the same number of carbon atoms, as they have an -OH group, which can form hydrogen bonds with other alcohol molecules. Hydrogen bonds are the strongest of the intermolecular forces.

Question 34(c)

Criteria	Marks
• Demonstrates a clear understanding of why the boiling points in a homologous series increase with increasing molar mass	1

Sample Answer

Larger molecules have more electrons present to allow for momentary dipoles to form and create stronger dispersion forces between molecules.

Question 35(a)

Criteria	Marks
• Provides the correct expression for the equilibrium constant	1

Sample Answer

$$K_{eq} = \frac{[SO_2Cl_2]}{[SO_2]\,[Cl_2]}$$

Question 35(b)

Criteria	Marks
• Correctly calculates the value of the equilibrium constant at 375°C, including: - determines initial concentrations of reactants - completes ICE table with final concentrations of reactants and products - calculates K_{eq} from ICE table values	3
• Provides some steps in the calculation	2
• Provides some relevant information	1

Sample Answer

	$[SO_2]$	$[Cl_2]$	$[SO_2Cl_2]$
initial	9.80×10^{-3}	9.80×10^{-3}	0
change	$- 9.75 \times 10^{-4}$	$- 9.75 \times 10^{-4}$	$+ 9.75 \times 10^{-4}$
equilibrium	8.83×10^{-3}	8.83×10^{-3}	9.75×10^{-4}

$$K_{eq} = \frac{9.75 \times 10^{-4}}{(8.83 \times 10^{-3})^2}$$

$$= 12.5$$

Appendix: Additional information for teachers

Question 1

ICE table

	[CO]	[Cl_2]	[$COCl_2$]
initial	2.0	2.0	0
change	- 1.885	- 1.885	1.885
equilibrium	0.115	0.115	1.885

$$K_{eq} = \frac{[\,COCl_2\,]}{[\,CO\,]\,[\,Cl_2\,]} = \frac{1.885}{(0.115)(0.115)} = 142.5$$

Question 4

$K_{sp} = [Pb^{2+}][Cl^-]^2$, let molar solubility = x

$K_{sp} = (2x)^2 \times (x) = 4x^3$

$x = \sqrt[3]{1.70 \times 10^{-5} / 4} = 0.0162 \text{ mol.L}^{-1}$

solubility = ($0.0162 \text{ mol.L}^{-1} \times 278.1 \text{ g.mol}^{-1}$) / 10 = 4.5×10^{-1} g in 100 mL

Question 9

$K_a = 10^{-4.2} = 6.3 \times 10^{-5} = [\,H^+\,][\,A^-\,] / [\,HA\,]$

$[H^+] = \sqrt{6.3 \times 10^{-5} \times 0.00023} = 1.2 \times 10^{-4}$

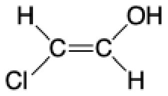
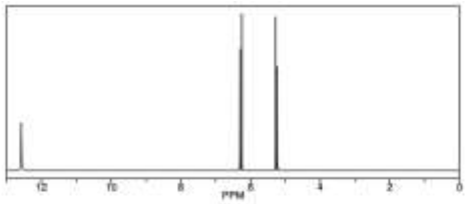
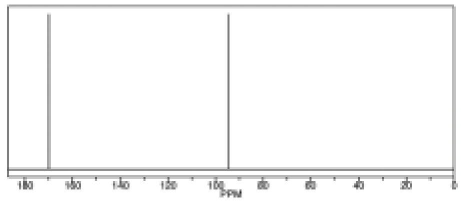
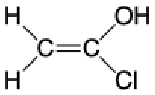
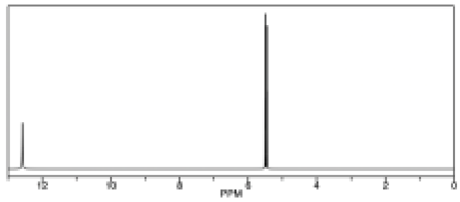
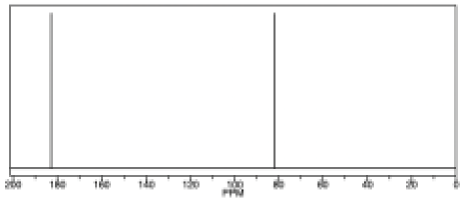
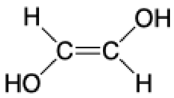
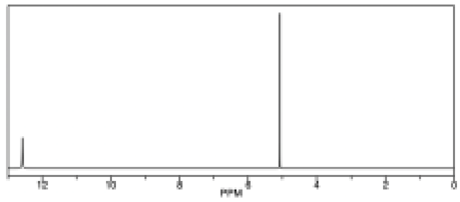
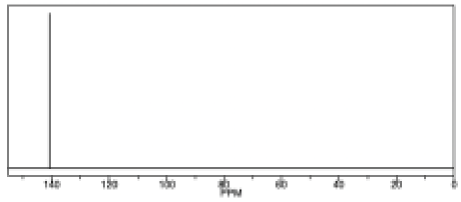
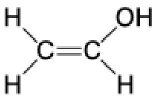
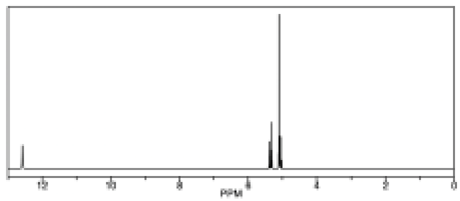
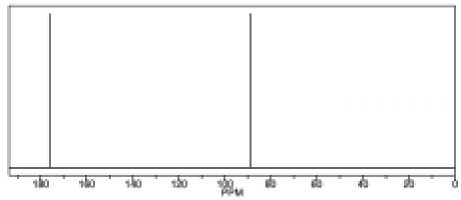
pH = $-\log(1.2 \times 10^{-4}) = 3.9$

Question 15

Mass of each ester subunit = (11 x 12.01) + (18 x 1.008) + (4 x 16.00) = 214.254 amu

Number of ester subunits = 5800 / 214.254 = 27. Hence 27 diol monomers incorporated.

Question 18

	Structure	^{1}H-NMR	^{13}C-NMR
A.			
B.			
C.			
D.			

Question 20

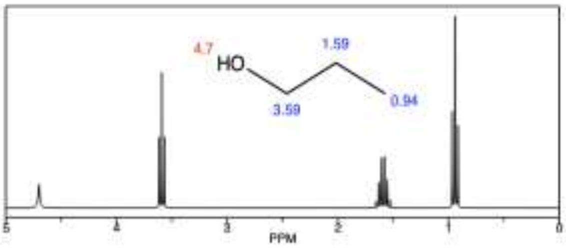
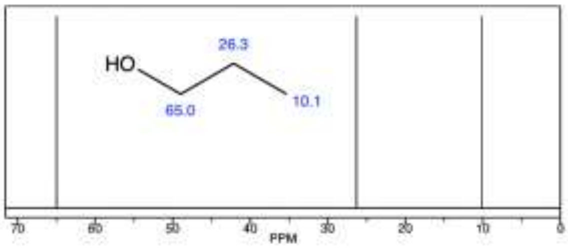
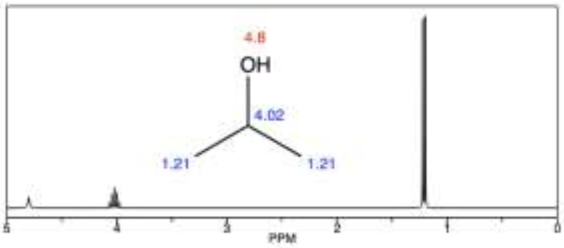
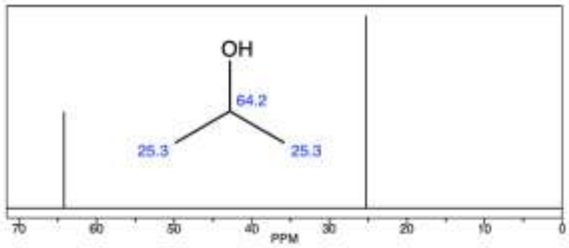
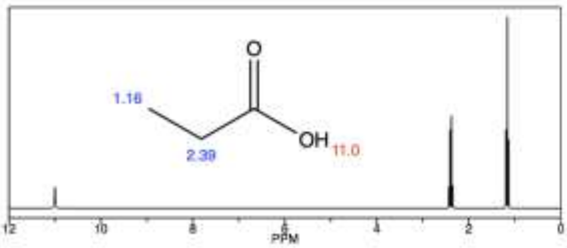
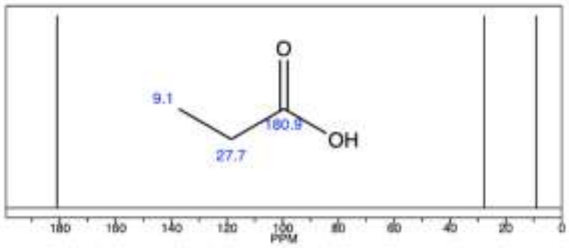
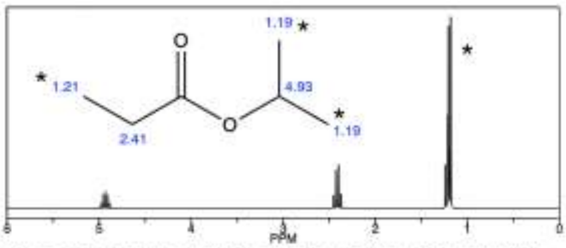
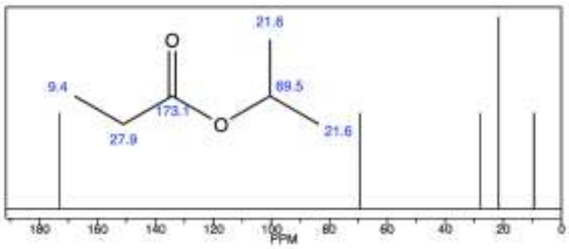
mol of $CaC_2O_4.H_2O$ = 2.86 g / 146.116 $g.mol^{-1}$ = 0.0196 mol

0.0196 mol of Ca = 0.0196 mol x 40.08 $g.mol^{-1}$ = 0.786 g

% of Ca in the phosphorite = 0.786 g / 4.64 g x 100% = 16.9%

Question 30

Chemical shift values for ^{1}H-NMR and ^{13}C-NMR spectra of B, C, D and E.

	H-NMR	C-NMR
B		
C		
D		
E	 * One complex signal at 1.2 as nine methyl H show overlap.	

Estimation quality is indicated by colour: good, medium, rough

Question 32(a)

$SrCO_3$ is less soluble than $BaCO_3$ so has not been included for the purposes of this question. The molar solubilities in the table were calculated from the K_{sp} values in the HSC Data Sheet and may vary from values found online.